

Synapse Database Diagram

First-pass relational model for Synapse, centered on courses, weighted assignments, grade history, syllabus imports, study activity, and a persistent academic graph.

Relationships

From	To	Relationship
USER	SEMESTER	owns
USER	COURSE	owns
USER	STUDY_SESSION	tracks
USER	WEEKLY_DIGEST	receives
SEMESTER	COURSE	contains
COURSE	ASSIGNMENT	has
COURSE	TOPIC	covers
COURSE	STUDY_SESSION	studied_for
COURSE	SYLLABUS_IMPORT	imported_from
ASSIGNMENT	GRADE	records
COURSE	PREREQUISITE	source_course
COURSE	PREREQUISITE	target_course

Entities

USER id uuid PK email text name text created_at timestamp updated_at timestamp	SEMESTER id uuid PK user_id uuid FK name text year int term text starts_on date ends_on date
COURSE id uuid PK user_id uuid FK semester_id uuid FK code text name text instructor text credits numeric syllabus_text text created_at timestamp updated_at timestamp	ASSIGNMENT id uuid PK course_id uuid FK name text category text weight numeric due_date date max_score numeric actual_score numeric status text created_at timestamp updated_at timestamp
GRADE id uuid PK assignment_id uuid FK score numeric letter_grade text date_entered timestamp	TOPIC id uuid PK course_id uuid FK name text description text confidence numeric
PREREQUISITE id uuid PK source_course_id uuid FK target_course_id uuid FK relationship_type text source text confidence numeric	STUDY_SESSION id uuid PK user_id uuid FK course_id uuid FK topic text duration_minutes int started_at timestamp ended_at timestamp

<i>SYLLABUS_IMPORT</i>	<i>WEEKLY_DIGEST</i>
id uuid PK	id uuid PK
course_id uuid FK	user_id uuid FK
file_name text	week_start date
raw_text text	week_end date
extracted_data json	summary text
status text	workload_snapshot json
created_at timestamp	grade_snapshot json
	created_at timestamp

Implementation Notes

- ? course is the central academic unit.
- ? assignment stores both deadlines and grading weights.
- ? grade preserves score history for trend analysis.
- ? topic and prerequisite form the academic graph.
- ? syllabus_import stores raw and structured extraction output for review and re-processing.
- ? study_session and weekly_digest support later workload forecasting and personalized summaries.